

TASK 1

Q1

```
#include <bits/stdc++.h>

using namespace std;

int secondLargest(vector<int> &a, int n) {

    int largest = a[0];

    int slargest = -1;


    for(int i = 1; i < n; i++) {

        if(a[i] > largest) {

            slargest = largest;

            largest = a[i];

        }

        else if(a[i] < largest && a[i] > slargest) {

            slargest = a[i];

        }

    }


    return slargest;

}
```

Q5

```
SELECT

    d.dept_name,

    e.name,

    e.salary

FROM Employees e

JOIN Departments d

    ON e.dept_id = d.dept_id
```

```
WHERE e.salary = (  
    SELECT MAX(e2.salary)  
    FROM Employees e2  
    WHERE e2.dept_id = e.dept_id  
);
```

Q6

```
SELECT  
  
    MONTHNAME(order_date) AS month,  
    SUM(amount) AS total_revenue,  
    COUNT(*) AS total_orders  
FROM Orders  
WHERE YEAR(order_date) = 2024  
GROUP BY MONTH(order_date), MONTHNAME(order_date)  
ORDER BY total_revenue DESC;
```

Q7

```
async function fetchUserPosts() {  
    try {  
        const response = await fetch(  
            "https://jsonplaceholder.typicode.com/posts?userId=3"  
        );  
  
        if (!response.ok) {  
            throw new Error(`HTTP Error: ${response.status}`);  
        }  
  
        const posts = await response.json();  
  
        posts.forEach(post => {
```

```

        console.log(post.title.toUpperCase());
    });

    console.log(`Total posts: ${posts.length}`);
} catch (error) {
    console.error("Failed to fetch posts:", error.message);
}
}

```

```

fetchUserPosts();

```

Q8

```

async function getWeather() {
    const city = "Delhi";
    const latitude = 28.6139;
    const longitude = 77.2090;

    const weatherCodes = {
        0: "Clear sky",
        1: "Mainly clear",
        2: "Partly cloudy",
        3: "Overcast",
        45: "Fog",
        61: "Light rain",
        80: "Rain showers",
        95: "Thunderstorm"
    };

    try {
        const url = `https://api.open-
meteo.com/v1/forecast?latitude=${latitude}&longitude=${longitude}&current_weather=true`;

```

```
const response = await fetch(url);

if (!response.ok) {
  throw new Error(`HTTP Error: ${response.status}`);
}

const data = await response.json();

const temperature = data.current_weather.temperature;
const weatherCode = data.current_weather.weathercode;

console.log(`City    : ${city}`);
console.log(`Temperature: ${temperature}°C`);
console.log(`Condition : ${weatherCodes[weatherCode] || "Unknown"}`);
} catch (error) {
  console.error("Failed to fetch weather data:", error.message);
}

}
```



```
getWeather();
```